

Confidence BER (conf BER) Algorithm

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Agenda

- Conf BER Algorithm

Simulation config: Parameters setting to obtain BER scale co-efficient for the data rate of 212.5Gbps PAM4

- num_bits: 3e6 (Number of bits, NBIT = 1.5e6, NBIT is the number of symbol)
- num_err: 0 (Number of errors)
- conf_lvl : 0.95 (95% confidence rate)
- target_ber: 1e-11; (Choose BER)
- Fb_Gbaud = 106.25 (Select Baud rate);
- sig_lvls = 4 (To select PAM4 signaling);

How to Calculate ber_conf, ber_sim and ber_scale?

- Calculate ber_conf

- ber_conf = 0.95 under ber ($ber_conf = abs(1 - conf_lvl - e^{-num_bits * ber} * \sum_{k=0}^{num_err} \frac{(num_bits * ber)^k}{k!})$)

- Calculate ber_sim

- ber_sim = 9.98584E-07
 - num_err is set to 0 in the model

$$ber_sim = fminsearch(ber_conf, \frac{num_err + 1}{num_bits})$$

- Calculate ber_scale

- ber_scale = 1.4193

$$ber_scale = \frac{sqrt(2) * erfcinv(sig_lvls * \frac{\log_2^{sig_lvls}}{sig_lvls-1} * target_ber)}{sqrt(2) * erfcinv(sig_lvls * \frac{\log_2^{sig_lvls}}{sig_lvls-1} * ber_sim)}$$

How to Calculate scaling factor, ber_c and ber_conf_raw?

- Scaling_factor:

- scaling_factor = 0.375 under PAM4

$$scaling_factor = \frac{sig_lvls - 1}{sig_lvls} * \frac{10}{floor(10 * \log_2^{sig_lvls})}$$

- ber_c

- err_sum is the simulation errors, which can be calculated after simulation

$$ber_c = \frac{err_sum + 1}{num_bits}$$

- ber_conf_raw

$$ber_conf_raw(ber_c) = abs(-1 * num_bits - \frac{\ln^{(1-conf_lvl)}}{ber_c} + \frac{\ln \sum_{k=0}^{err_sum} \frac{(num_bits * ber_c)^k}{k!}}{ber_c})$$

How to Calculate ber_conf_sim_raw and ber_conf?

- ber_conf_sim_raw

$$ber_conf_sim_raw = fminsearch(ber_conf_raw, \frac{err_sum + 1}{num_bits})$$

- ber_conf

$$ber_conf = scaling_factor * erfc(erfcinv(\frac{1}{scaling_factor} * ber_conf_sim_raw) * ber_scale)$$

- SNPS pass-fail criteria
 - MLSD Conf BER should be lower than 1e-6



Thank you